

Individual Test

MINNESOTA YOUTH MATH OUTREACH

August 31, 2024

- The individual round is an **hour-long (60-minute)** contest covering all four topic areas taught during the duration of the 2024 MNYMO camp.
- There are **20 problems** in this round, scaling higher in difficulty for each problem.
- Each problem is worth 1 point and there is **NO** penalty for guessing.
- Answer submissions are only considered valid if they are written on the dedicated answer individual test sheet you are given. The problem sheet will not be collected and therefore no work in the problem sheet itself will be evaluated.
- **In case of a tie**, a final tiebreak problem will be utilized. **Problem 21** is meant to be an estimation and the submission with the closest numerical value to the answer on a logarithmic scale will be given the higher rank (Specifically for the tiebreaker, if the correct answer is A and the submitted answer is N , the lower value of $|\log(\frac{N}{A})|$ wins). **You will lose all ties automatically if you don't put an answer**, so be sure to submit something for this problem!

Ranking of individual scores will proceed as follows:

1. Higher individual test score (problems 1 to 20).
2. Closest tiebreak estimation (problem 21).
3. Latest last problem solved (For example, solving P20 would take precedence over P19).
4. Latest second last problem solved (This will continue to iterate following steps 3 and 4).

Problem 1. Jinu Lee is going on a diet. He goes to the local fruit market and notices that three apples and four oranges together cost \$12, while four apples and two oranges together cost \$11. How many cents does one apple cost?

Problem 2. Find the area of a rectangle with one side of length 4 and diagonals of length 5.

Problem 3. Ben is finding the square root of a three digit perfect square x . In doing so, however, he accidentally swaps the tens and units digits of x and gets an answer that is one more than the correct answer. Find x .

Problem 4. How many of the first 200 positive integers are multiples of 5 but not 6?

Problem 5. Sam, Gary, and Owen decide to share fruits. Sam has 50 strawberries, Gary has 40 grapes, and Owen has 30 oranges to begin with. Given that any type of fruit can only be given in groups of 6, and every person has all 3 types of fruit at the end, what is the maximum number of fruits Gary could have at the end of the exchange? (Note: Not all the fruit someone initially has must be given out.)

Problem 6. Find the sum of the prime factors of the number $S = 4^8 - 1$.

Problem 7. How many ways are there to color the vertices of a tetrahedron (a 3D figure made up of 4 vertices, 4 triangular faces, and 6 edges), where each vertex can be colored with one of six colors, such that vertices connected by an edge have distinct colors? Colorings that differ by rotations and/or reflections are considered distinct.

Problem 8. Alexander Ren and Sam Kretzschmar are both listening to the same song on repeat. However, Sam accidentally set it to play at 1.25 times speed, so for every one second he listens to the song, 1.25 seconds of the original song has elapsed. Given that the song is 60 seconds long and they both start playing it at the same time, how many seconds does it take for Sam and Alex to start playing the song at the same time again?

Problem 9. Integers x and y satisfy the equation $xy = x - y + 5$. Find the greatest possible value of $x + y$.

Problem 10. Four spheres of radius 1 are arranged such that their centers form the vertices of a square of side length 2. A larger sphere is placed around these four spheres such that the larger sphere is internally tangent to each smaller sphere. Find the smallest possible value of the radius of the larger sphere.

Problem 11. An integer k is considered to be *level* n if k ends in exactly n 0's. For example, the number 100 is level 2. What is the smallest integer k such that $k!$ is level 10?

Problem 12. Find the sum of all real numbers m such that the polynomial $P(x) = x^2 + mx + m + 2024$ has exactly one distinct root.

Problem 13. Call a number *vanilla* if at least half of its digits are the same, and call a number *unique* if all of its digits are distinct from each other. What is the ratio of the number of vanilla 4-digit numbers to the number of unique 4-digit numbers?

Problem 14. Ricky flips a fair coin five times, and Miguel flips a fair coin three times. What is the probability that Ricky lands more heads than Miguel?

Problem 15. Consider the polynomial $p(x) = x^{100} + 2x^{99} + \dots + 100x + 101$. Let the roots of $p(x)$ be $r_1, r_2, \dots, r_{100}$. Compute $\sum_{i=1}^{100} \frac{1}{r_i}$.

Problem 16. Let ABC be a triangle, and let D and E be points on sides BC and AC , respectively, such that $BD : DC = 1 : 2$ and $AE : EC = 1 : 3$. Let lines $\overline{AD}$ and $\overline{BE}$ intersect at P , and let line $\overline{CP}$ intersect side AB at point F . Compute CP/PF .

Problem 17. In each of its moves, the *downright frog* moves from its current point (x, y) to $(x + 1, y - 2)$ or to $(x + 2, y - 1)$. Compute the number of ways for the downright frog to move from $(0, 15)$ to $(15, 0)$.

Problem 18. Points A, B, C, D , and E lie on a circle with radius 5, in that order. Given that $\angle ABC = 90^\circ$, $CD = 6$, and $\cos(\angle ABE) = \frac{2}{3}$, the length of DE can be written in the form $\frac{a-b\sqrt{c}}{d}$, where a, b, c , and d are positive integers such that c is squarefree and $\gcd(a, b, d) = 1$. What is $a + b + c + d$?

Problem 19. The real number x satisfies the equation $x + \frac{1}{x} = 3$. The value of $x^{2024} + \frac{1}{x^{2024}}$ can be written as $\frac{m}{2^{2024}}$. Find the remainder when m is divided by 9.

Problem 20. A group of 16 people stand equally spaced in a circle, facing inwards. Alice and Bob stand opposite one another in the circle, and Bob is handed a ball. Every second, the person holding the ball gives it to the person to their left with probability $\frac{2}{5}$ and the person to their right with probability $\frac{3}{5}$. If the probability that Alice is first given the ball from the person to her right is $\frac{m}{n}$, for relatively prime positive integers m and n , find $m + n$.

Problem 21 (Tiebreaker). Define a *limited number* to be a positive integer which has exactly two distinct digits in its decimal representation (leading zeros are not considered). For example, 3344433 and 20 are limited numbers, but 0111 and 12332 are not. Find the 225th limited number which is divisible by 90. If the answer is A and your solution is N , the lowest value of $\left\lceil \log \left(\frac{N}{A} \right) \right\rceil$ wins.